

Special instructions

- This exam consists of **three parts**. The mark breakdown is as follows:

Part	A	B	C			
Question	1	2	3	4	5	Total
Marks	15	15	10	10	10	60

- **Unless a question indicates a numeric value is required**, numeric answers may be left as numeric expressions if desired (e.g., if the correct answer was 2, an answer of ' $1 + 1$ ' would receive the same credit as an answer of '2'). If you express answers as numeric values these should be accurate to at least three decimal places.
- If you are asked to identify the type of a variate, choose from Continuous, Discrete, Categorical, Ordinal, or Complex.
- If you believe insufficient information is provided in a question to determine the requested answer, you may write 'not enough information', 'NEI', or similar. If the question indicates you should justify your answer, this justification is still required even if your answer is 'not enough information'.
- Additional blank pages are provided at the end of the exam for rough work and/or additional space for answers. If you wish any work on these pages to be graded, you must clearly indicate this on the corresponding question page.
- Good luck :)

Part A [15 marks]

Question 1 [15 marks]

(a) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) Which of these numerical summaries is a measure of shape?

- A The sample standard deviation
- B The interquartile range
- C The sample skewness
- D The sample variance

(ii) Which of the following can always be determined from a relative frequency histogram of a sample of data?

- A The sample size
- B The sample mode
- C The sample standard deviation
- D None of the above

(iii) Which probability distribution would be most appropriate to model a continuous variate?

- A Multinomial
- B Binomial
- C Poisson
- D Exponential

(iv) The sample correlation may be an appropriate numerical summary for what kind of data?

- A Univariate data
- B Bivariate data
- C Both univariate and bivariate data
- D Neither univariate nor bivariate data

(v) If $R(\theta)$ is the relative likelihood function and $\hat{\theta}$ the maximum likelihood estimate, then what does $R(\hat{\theta})$ equal?

- A 0
- B 1
- C It depends on the value of $\hat{\theta}$
- D It depends on the value of θ

(b) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. You are not required to show any working for your calculations. No part marks are available.

A sample of size $n = 5$ contains the values $-4, -2, 0, 2, 4$. These data are stored in R in an object named `x`.

(i) What is the sample mean of `x`? _____

(ii) Is the sample standard deviation of `x` positive, negative, or zero? _____

(iii) Is the sample skewness of `x` positive, negative, or zero? _____

(iv) What number would be output if you ran the command `sum(x)`?

(v) Give *one* reason why it is impossible these data were sampled from a Poisson distribution.

(c) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. You are not required to show any working for your calculations. No part marks are available.

Suppose $y_1, y_2, \dots, y_{50}$ is an observed random sample from a probability distribution with corresponding independent and identically distributed random variables $Y_1, Y_2, \dots, Y_{50}$. You are told that the minimum value in the sample is 1, the maximum value in the sample is 10, and that:

$$\sum_{i=1}^{50} y_i = 100 \quad \sum_{i=1}^{50} (y_i - \bar{y})^2 = 600$$

where $\bar{y}$ denotes the sample mean.

(i) Suppose $y_1, y_2, \dots, y_{50}$ is an observed random sample from a $G(\mu, \sigma)$ distribution. What is the maximum likelihood estimate of μ ?

(ii) Suppose that $Y_1, Y_2, \dots, Y_{50} \sim G(\mu, \sigma)$, independently. What is the maximum likelihood estimate of σ^2 ?

(iii) Suppose that $Y_1, Y_2, \dots, Y_{50} \sim \text{Poisson}(\theta)$, independently. What is the maximum likelihood estimate of θ ?

(iv) Suppose that $Y_1, Y_2, \dots, Y_{50} \sim \text{Poisson}(\theta)$, independently. What is the maximum likelihood estimate of the standard deviation of Y_5 ?

(v) Suppose that $Y_1, Y_2, \dots, Y_{50} \sim \text{Poisson}(\theta)$, independently. What is the maximum likelihood estimate of $E \left[\sum_{i=11}^{20} Y_i \right]$? Hint: $\sum_{i=1}^n Y_i \sim \text{Poisson}(n\theta)$.

Part B [15 marks]

Part B begins on the following page. All questions in Part B are based on Assignment 1. All variate names refer to those in the Assignment Dataset Information file. All R commands assume the assignment dataset has been stored in R in an object named `mydata`. You may further assume that the variate `subject.age.log` has been created (as a variate within `mydata`) as described in Analysis 2 of Assignment 1.

Important: Unless otherwise stated, you should assume that a question is asking about the entire sample, and not one specific city.

Question 2 [15 marks]

(a) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) What type of variate is `vehicle.colour`?

- A Continuous
- B Discrete
- C Categorical
- D Ordinal

(ii) Which of these R commands would generate a scatterplot with `subject.age` as the response variate and `time.hour` as the explanatory variate for the assignment dataset?

- A `plot(mydata$subject.age, mydata$time.hour)`
- B `plot(mydata$time.hour, mydata$subject.age)`
- C Both A and B could be used.
- D Neither A nor B could be used.

(iii) Which of these R functions should be used when drawing a Gaussian probability density function curve on a relative frequency histogram?

- A `dnorm()`
- B `pnorm()`
- C `rnorm()`
- D `qnorm()`

(iv) How many traffic stops in the dataset in New Orleans happened on a Monday?

- A 12.5
- B 49
- C 349
- D 634

(v) What is the output of running the command

```
sum(mydata$subject.age.log)
```

- A [1] 3.495931
- B [1] 35.25895
- C [1] 3321.135
- D [1] 33496

(b) [5 marks] Analysis 2c in Assignment 1 was the following:

- 2c. Generate an empirical cumulative distribution function plot with a suitable superimposed Gaussian cumulative distribution function curve for each of `subject.age` and `subject.age.log` for San Antonio. Your Gaussian cumulative distribution function curves should be a different colour to the e.c.d.f.

A student attempts to answer this question. You may assume they have correctly created an object named `SanAntonio` that contains only the traffic stops from San Antonio. The student then used the following R commands to answer Analysis 2c:

```
> mu <- median(SanAntonio$subject.age)
> sigma <- sd(SanAntonio$subject.age)
> plot(ecdf(SanAntonio$subject.age), xlab = "subject.age.log", main = "ecdf")
> curve(dnorm(x, mu, sigma))
```

Identify five mistakes the student has made in their R code. For each mistake, you should briefly describe what problem would arise in the resulting output. You do *not* need to explain how the code could or should be corrected, but you may do so if you feel it aids in your explanation. Note that simply providing a corrected R command is not sufficient to receive full credit: your answer must describe why the preceding code does not produce the desired output. Note: The plot's title is simply 'ecdf', this is *not* a mistake for the purposes of this question.

(c) [5 marks] Consider each of the following statements based on the assignment. For each statement, discuss whether you agree or disagree, providing a brief justification for your answer in 1-2 sentences.

(i) “Based on the sample correlation there is no relationship between `subject.age` and `vehicle.year` in San Antonio in the assignment dataset.”

(ii) “Because the sample mean of `subject.age` is different to the sample median of `subject.age` we can conclude that a Gaussian model is not appropriate for `subject.age` in San Antonio.”

(iii) “The median vehicle colour in San Antonio in the assignment dataset is white.”

Part C

Question 3 [10 marks]

Alex and Blake are two students working on co-op for *Goose River Transit*, a public transit company that provides bus services to residents of an area called Goose River. 12 months ago, the company introduced the *FlapPass*, a digital payment method that can be used instead of cash. Passengers may still pay in cash if they wish, but the cash fare is \$4 per journey, while each journey costs \$3 using the FlapPass.

The company conducted a survey of passengers while they were waiting for buses at the Goose River Transit Centre. A total of 100 passengers provided a response to the survey. Alex and Blake have been asked to analyze the resulting data, a summary of which is given in the table below.

Variate	Description	Sample Statistics	
Age	The passenger's age, in years.	Mean: 50.0	SD: 21.5
Journeys	The number of journeys the passenger took in the previous seven days.	Range: 0-28 Median: 12	Mean: 12.3 SD: 4.0
FlapPass	Whether the customer owns a FlapPass ('yes' or 'no').	Yes: 50	No: 50

(a) [2 marks] Based on the preceding information, answer each of the following:

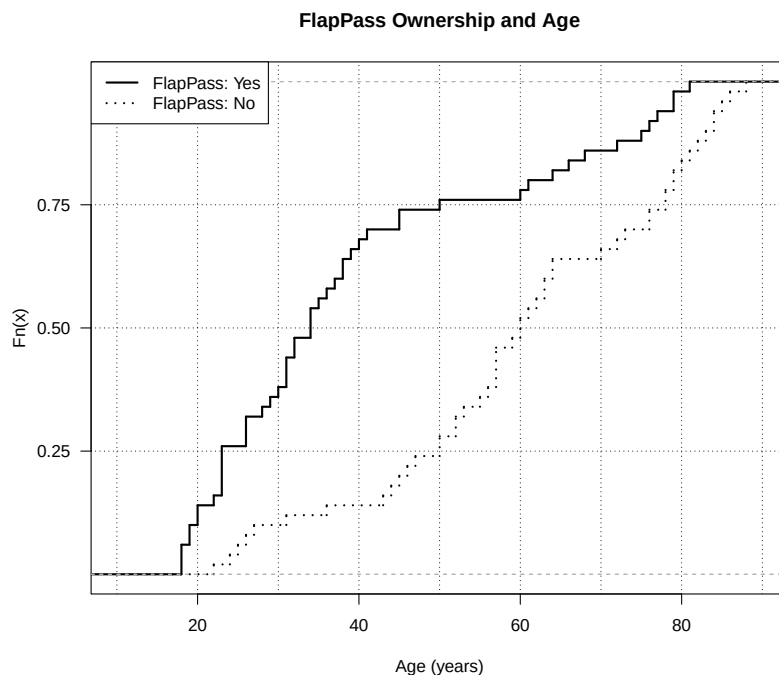
(i) What type of variate is FlapPass? _____

(ii) What is the interquartile range of Journeys in the sample? _____

Let $j_1, j_2, \dots, j_{100}$ denote the number of journeys taken by each passenger in the sample. Assume these are realizations of a random sample $J_1, J_2, \dots, J_{100}$ where $J_i \sim \text{Poisson}(\theta)$, $i = 1, 2, \dots, 100$, independently.

(b) [3 marks] Give a numeric expression for the maximum likelihood estimate of the probability a randomly chosen passenger from the population took at least 14 journeys in the past seven days. Clearly indicate any results you use from the Course Notes.

The plot below contains the e.c.d.f. for passenger age, split between those who have a FlapPass and those who do not. Horizontal and vertical dotted lines have been added to aid readability.



(c) [3 marks] *Estimate* a numeric value for each of the following. Your answer may be within ± 3 years of the correct value, except for part (iii) where your answer may be within ± 5 years of the correct value. Note: ‘not enough information’ is not a valid answer to any of these questions, even if you cannot calculate an exact answer, you should still estimate one.

(i) The age of the oldest passenger in the sample. _____

(ii) The median age of passengers in the sample who do not own a FlapPass. _____

(iii) The interquartile range of age of passengers in the sample who own a FlapPass. _____

(d) [2 marks] Blake tells Alex that because an equal number of people in the sample do and do not own a FlapPass, the mean age of those in the sample who own a FlapPass must be the same as those who do not own a FlapPass. Do you agree with Blake? Briefly explain why or why not.

Question 4 [10 marks]

Charlie is the owner of a bubble tea and video game café called *MOBA Boba*. For several years the café has run a promotion on Tuesdays called *BOGO Boba* where customers can get two bubble teas for the price of one. Earlier this month, Charlie decided they wanted to study how long customers spend at the café, and whether they spend longer at the café during the *BOGO Boba* promotion.

Charlie took a sample of 30 customers from Monday September 8 and 30 customers from Tuesday September 16, and recorded the length of their visit rounded to the nearest second. Note that these were all the customers that visited the café on those days.

(a) [2 marks] Answer the following questions about the study.

(i) Is this an observational study or an experimental study?

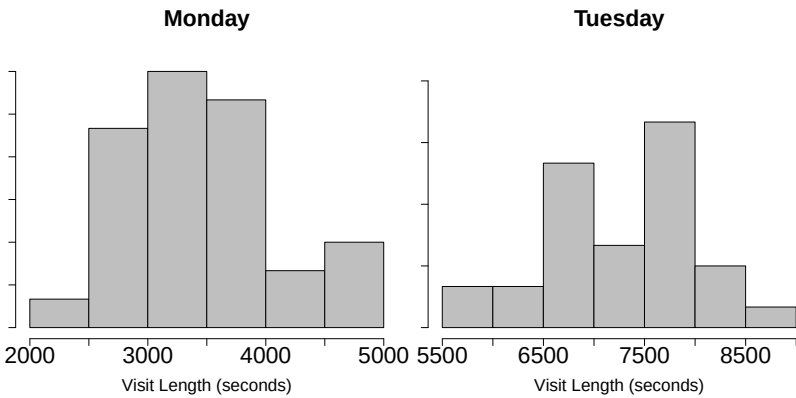
(ii) Carefully define the units in the study.

Let $v_1, v_2, \dots, v_{60}$ denote the customer visit times in Charlie's sample. Assume these are realizations of a random sample $V_1, V_2, \dots, V_{60}$ where $V_i \sim G(\mu, \sigma)$, $i = 1, 2, \dots, 60$, independently.

(b) [2 marks] Based on the information given in the question above, give one reason why the proposed model is not appropriate. Your answer should include a brief justification.

(c) [2 marks] Charlie explains to one of their employees that they have concerns about using a Gaussian model for these data. The employee suggests converting the times from seconds to hours by dividing by 3600 to try and improve the suitability of the Gaussian model. Do you agree or disagree with this advice? Justify your answer.

Charlie constructed relative frequency histograms of the visit times for the two days, as shown in the following figure. Note that the y -axis labels have been omitted.



(d) [2 marks] Based on the figure, for each of the following statements indicate whether it is True, False, or if Not Enough Information (NEI) is provided to determine if it is true or false.

Question	T/F/NEI
(i) The sample mean visit time on Tuesday was lower than the sample mean visit time on Monday.	
(ii) The sample median visit time across the entire sample of 60 customers was less than 6000 seconds.	
(iii) The range of visit times on Tuesday was higher than the range of visit times on Monday.	
(iv) The modal visit time was between 3000 and 3500 seconds.	

(e) [2 marks] Let s_m denote the sample standard deviation of visits on Monday, s_t denote the sample standard deviation of visits on Tuesday, and s_v denote the sample standard deviation of visits across the entire 60-visit sample. Which of the following statements is definitely true? Carefully justify your answer. If you believe it is not possible to determine if any of these statements are definitely true, then say so (with justification) in your answer.

$$s_v > s_m \qquad s_v > s_t \qquad s_v < s_m \qquad s_v < s_t$$

Question 5 [10 marks]

A casino has a game which costs \$10 to play. Each play can result in one of three outcomes with an associated reward: a loss (\$0); a small prize (\$10); or a large prize (\$100). The casino states that the probability of winning a small prize is five times the probability of winning a large prize.

Let the random variable W represent the outcome of a random play of the game. We will state that W can take on the values 0, 10, or 100 (corresponding to losing, a small prize, or a large prize, respectively). We can write the probability function for W as follows:

$$P(W = w; \theta) = \begin{cases} 1 - 6\theta & w = 0 \\ 5\theta & w = 10 \\ \theta & w = 100 \end{cases}$$

and 0 otherwise. Note that the range of θ has been deliberately omitted.

You go to the casino and play the game 10 times. You win the large prize once, the small prize 4 times, and lose the remaining 5 times.

(a) [1 mark] Write down the possible range of θ .

(b) [4 marks] Find the maximum likelihood estimate of θ based on these data. You may assume all plays of the game are independent.

(c) [2 marks] What is the maximum likelihood estimate of the expected profit or loss for a single play of the game? You may write your answer in terms of $\hat{\theta}$. Hint: Begin by calculating $E[W]$. You are reminded the game costs \$10 to play.

Part (d) concerns a different probability distribution, unrelated to parts (a)-(c).

Let $z_1, z_2, \dots, z_n$ be independent, identically distributed observations drawn from a continuous probability distribution with parameter θ . You are told that the probability density function for this distribution is:

$$g(z; \theta) = \theta z e^{z/\theta} \quad z > 0 \text{ and } \theta > 0$$

Note: This is not a real probability density function, but you may assume it has all the properties necessary in order to solve the following questions, and as such you may answer the following under the assumption it is a real probability density function. In particular, you may assume that the likelihood and log-likelihood functions are maximized for the value of θ at which their first derivatives equals 0.

(d) [3 marks] Find the maximum likelihood estimate of θ , expressed as a function of the observed data. Carefully show all your steps.

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